QF620 Stochastic Modelling in Finance — Session 6 Reference Cheatsheet

Valuation Framework and Stochastic Volatility Models: pricing vs reporting models, implied volatility and the smile, displaced diffusion, stochastic volatility, Heston, SABR

Built from Session 6 slides (Tee Chyng Wen, SMU) · October 2026 · revised edition with intuition and full slide derivations

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Source: "Session 6: Valuation Framework and Stochastic Volatility Models" (Tee Chyng Wen, SMU, 20 slides).

How to read this sheet. Each concept entry has up to five parts: **The idea in words** (what the concept is and why it is there), the formula with a shaded where box naming every symbol, **Derivation** (the slide's argument line by line, with the reason for each step in brown), **Worked example**, and **Exam angle** (traps

and what to be able to reproduce). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △! marks a slide ambiguity or error, with the corrected version.

What you must be able to do after this session: distinguish pricing and reporting models; define implied volatility and explain the smile and its history; explain why one-parameter models cannot fit the smile; solve the displaced-diffusion SDE and price it with the Black formula (and know its $\beta \to 0, 1$ limits); solve the lognormal volatility SDE; write the Heston and SABR SDEs and say what each parameter does.

Notation used throughout (slide conventions)

Symbol	Meaning	Notes
$S,\ F_t,\ F_0$	spot; forward price (driftless under the pricing measure); today's forward	[S10]
$K,\ T,\ r$	strike; expiry; rate	
σ_K	implied volatility at strike K	[S4]
$\sigma_{LN},\ \sigma_N$	lognormal (Black–Scholes) / normal (Bachelier) volatility	[S8]
β	displaced-diffusion mix (DD) or CEV exponent (SABR), $\in [0, 1]$	[S10, S17]
W_t^*	Brownian motion under the pricing measure	[S12]
ν	volatility of volatility	[S15]
$\sigma_t,\ \alpha_t$	stochastic volatility	[S15, S17]
V_t; κ, θ	instantaneous variance (Heston); reversion speed, long-run variance	[S16]
ρ	correlation between price and vol Brownian motions	[S16–S17]
σ_{SABR}	SABR (Black) implied volatility	[S18]
$\text{BlackCall}(F, K, \sigma, T)$	Black (forward) call formula	[S11]

Notation trap: β means two things: the DD weight on the lognormal part [S10] and the SABR backbone exponent F^β [S17]. In both, $\beta = 1$ is lognormal and $\beta = 0$ is normal. ν is vol-of-vol; V_t is a variance (Heston); σ_t, α_t are volatilities.

0. Topic map

```
PREREQUISITES: S5 Bachelier and –BlackScholes calls, solving SDEs with Itô ; S4 dW dW = dt
    ▼
SESSION 6├─
 REPORTING [–S2S3]│
     ├─ pricing models: price & hedge ; realistic dynamics ; accurate Greeks    [S2]│
     └─ reporting models: market price ↔ implied vol, simple formula            [S3]├─
 SMILE [–S4S7]│
     ├─ implied vol σ_K: BS(S, K, r, σ_K, T) = market price                      [S4]│
     └─ flat before 1987, skewed after ; steeper for short expiries            [–S5S7]├─
 EXTENSIONS TO –BLACKSCHOLES [–S8S9]│
     └─ one parameter fits ATM only ; market between normal and lognormal       [–S8S9]├─
 DISPLACED DIFFUSION [–S10S14]│
     ├─ dF = σβ[F + –β(1)₀F]dW ; solution via log                              [S10]│
     ├─ price = Black(₀Fβ/, K + –β(1)₀Fβ/, σβ, T)                               [–S11S12]│
     └─ better fit, not exact ; returns show volatility clustering             [–S13S14]└─
 STOCHASTIC VOLATILITY / SABR [–S15S20]
      ├─ σd = νσdWσ^ ⇒ σ_T = σ₀exp–(ν½²T + νW_Tσ^)                                [S15]
      ├─ Heston: √V, mean-reverting variance, ρ                                   [S16]
      ├─ SABR: dF = αFβ^ dW^F, αd = να dWα^, ρ ; Hagan's σ_SABR → Black           [–S17S18]
      └─ DD lacks degrees of freedom ; SABR fits the smile                        [–S19S20]
```

The storyline of the session.

- Market option prices are set by supply and demand. The Black–Scholes formula is used *backwards*, as a reporting tool, to turn each price into an implied volatility.
- If Black–Scholes were right, implied vol would be flat across strikes. Since 1987 it is not (the **smile/skew**). So a pricing model needs more freedom than one σ.
- **Displaced diffusion** interpolates between normal and lognormal with one extra parameter β, and is still priced with the Black formula. It tilts the skew but cannot bend it.

- Real returns show **volatility clustering**, so make volatility itself random. Vol-of-vol ν adds curvature, correlation ρ adds skew. SABR has a closed-form implied vol, hence it is fast and popular; Heston needs a numerical integral.

1 Concept entries

1.1 Pricing models vs reporting models [S2–S3]

The idea in words
Two different jobs. A **pricing model** is what a desk uses to value and hedge its book: its dynamics should be realistic and its Greeks accurate, and its parameters are calibrated to liquid options. A **reporting model** is just a units converter: exchanges and brokers quote options as implied vols instead of prices, and need a simple closed-form formula (Black–Scholes or Bachelier) to convert one into the other. The reporting model does not need to be "true".

Pricing models [S2]: dynamics conform to intuition about the underlying; Greeks capture sensitivities accurately; each desk has its own, calibrated to liquid option markets.

Reporting models [S3]: report market prices (driven by supply and demand) as implied volatilities; make simplifying assumptions to get an analytic price–vol conversion; their parameters are displayed on brokers' screens.

Slide question [S3]: why quote vols, not prices? (answered here) Implied vol strips out spot, strike, expiry and rates, so options on different strikes, maturities and underlyings are comparable on one scale; it changes much less than the price when spot moves; and it is what traders actually have views on.

1.2 Implied volatility [S4]

The idea in words
Take the market price of one option and ask: which σ in Black–Scholes reproduces it? That number is the implied volatility. Each strike gets its own. It is solved numerically, but it always exists and is unique because the Black–Scholes price strictly increases with σ (more volatility = more optionality).

$$C^{\text{mkt}}(K_i) = \text{BlackScholes}(S, K_i, r, \sigma_{K_i}, T), \qquad i = 1, 2, 3, \ldots$$

where

$C^{\text{mkt}}(K_i)$	observed market price of the call (or put $P(K_i)$) with strike K_i.
σ_{K_i}	implied vol for strike K_i: solved numerically (no closed-form inverse); one per strike.

Why it is unique, and calls and puts agree *[Beyond slides]*
Vega $= \frac{\partial C}{\partial \sigma} = S\phi(d_1)\sqrt{T} > 0$, so the price is strictly increasing in σ and the root of $C(\sigma) = C^{\text{mkt}}$ is unique. By put–call parity $C - P = S - Ke^{-rT}$, which does not depend on σ; so if σ matches the call it also matches the put with the same strike and expiry.

1.3 The volatility smile [S5–S7]

The idea in words
Plot implied vol against strike. Under Black–Scholes it is a flat line. In equity markets after the 1987 crash it slopes down: low strikes (OTM puts) are priced at higher vols. Reason: the market discovered that 20% one-day crashes happen, so it pays up for downside protection. Equivalent statement: the market's risk-neutral distribution has a fatter left tail than the lognormal (negative skew, excess kurtosis — Session 1's γ_3, γ_4). Short expiries show the steepest smile because a sudden jump matters most over a short horizon.

- Black–Scholes assumes one constant volatility, independent of strike and expiry ⇒ flat implied vol [S5–S6].
- Before Black Monday (October 1987) the index smile was roughly flat; afterwards a distinct smile/skew appeared across many markets: a 20%+ drop in a short time was shown to be possible [S5–S6].

- Prices come from supply and demand, not a model; higher implied vols for low strikes reflect demand for protection against large down moves [S6].
- The smile is steepest for short expiries, flatter for long ones [S7]. The market trades mostly OTM and ATM options; ITM ones are less liquid, so for equity indices there is heavy demand for OTM puts [S7].

1.4 Fitting market prices with one-parameter models [S8–S9]

The idea in words
Black–Scholes and Bachelier each have one volatility number. Fit it to the most important option (ATM) and you are stuck with whatever shape the model implies elsewhere: Black–Scholes gives a flat line in implied vol; Bachelier gives a fixed downward slope. Why downward: a normal model has the same *absolute* move size at every level, so measured in *percentage* terms the move is bigger at low strikes, i.e. higher lognormal vol. The market sits between the two.

$$\text{model price}(K_i) = \text{BlackScholes}(S, K_i, \sigma_{LN}, T) \quad \text{or} \quad \text{Bachelier}(S, K_i, \sigma_N, T).$$

where

σ_{LN}	one lognormal (percentage) vol for all strikes.
σ_N	one normal (price-unit) vol for all strikes; to match ATM, $\sigma_N \approx \sigma_{LN} S$ *[Beyond slides]*.

Google example [S9]: options on 2013-08-30 expiring 2015-01-17 ($T \approx 1.38$), spot 846.9, ATM vol ≈ 0.26. Converted to Black implied vols: lognormal model flat at 0.26; normal model downward-sloping (*[Beyond slides]*: about 0.37 at $K = 400$, 0.26 ATM, 0.20 at $K = 1400$); market in between, with some upturn at high strikes.

Heuristic for the Bachelier skew *[Beyond slides]*
Local percentage volatility of $\mathrm{d}F = \sigma_N \,\mathrm{d}W$ at level F is σ_N / F. Near strike K the option "feels" the volatility around K, so roughly $\sigma_{\text{impl}}(K) \approx \sigma_N / \frac{F_0 + K}{2}$: larger for $K < F_0$, smaller for $K > F_0$.

1.5 Displaced-diffusion model: SDE and solution [S10]

The idea in words
Blend the two: the noise is a weighted mix of a lognormal piece ($\propto F_t$) and a normal piece (a constant, frozen at F_0). $\beta = 1$ is pure Black, $\beta = 0$ pure Bachelier; in between gives an intermediate skew. To solve it, notice that the quantity $\beta F_t + (1-\beta)F_0$ itself follows a GBM (its noise is proportional to itself), so take its log, exactly as for Black–Scholes. The result is a **shifted lognormal**: a lognormal variable minus a constant.

$$\mathrm{d}F_t = \sigma\big[\beta F_t + (1-\beta)F_0\big]\,\mathrm{d}W_t, \qquad \beta \in [0, 1].$$

where

F_t	forward price, driftless under the pricing measure; F_0 is a fixed constant.
βF_t	lognormal (geometric) part: noise proportional to the level.
$(1-\beta)F_0$	normal (arithmetic) part: noise of fixed size.
σ	at $t = 0$ the local volatility is σF_0, same as Black's.

Solving via Itô (the slide gives only the answer)
Let $Y_t = \beta F_t + (1-\beta)F_0$. Then $\mathrm{d}Y_t = \beta\,\mathrm{d}F_t = \beta\sigma Y_t\,\mathrm{d}W_t$: a driftless GBM with vol $\beta\sigma$. Take $X_t = f(F_t) = \log Y_t$, $f' = \frac{\beta}{Y}$, $f'' = -\frac{\beta^2}{Y^2}$, $\sigma_t = \sigma Y_t$, $\mu_t = 0$:

$$\begin{aligned}
\mathrm{d}X_t &= \Big(0 + \tfrac{1}{2}\sigma^2 Y_t^2 \cdot \Big(-\frac{\beta^2}{Y_t^2}\Big)\Big)\,\mathrm{d}t + \sigma Y_t \cdot \frac{\beta}{Y_t}\,\mathrm{d}W_t = -\tfrac{1}{2}\beta^2\sigma^2\,\mathrm{d}t + \beta\sigma\,\mathrm{d}W_t && \text{(Itô)}\\
X_T &= \log F_0 - \tfrac{1}{2}\beta^2\sigma^2 T + \beta\sigma W_T && (Y_0 = F_0)\\
\beta F_T + (1-\beta)F_0 &= F_0 \exp\Big(-\tfrac{1}{2}\beta^2\sigma^2 T + \beta\sigma W_T\Big) && \text{(exponentiate)}\\
F_T &= \frac{F_0}{\beta}\exp\Big(-\frac{\beta^2\sigma^2 T}{2} + \beta\sigma W_T\Big) - \frac{1-\beta}{\beta}F_0 && \text{(solve for } F_T)
\end{aligned}$$

Limits *[Beyond slides]*
$\beta = 1$: $F_T = F_0 e^{-\frac{1}{2}\sigma^2 T + \sigma W_T}$ (Black). $\beta \to 0$: expand $e^x \approx 1 + x$ with $x = \beta(\sigma W_T) + O(\beta^2)$: $F_T \approx \frac{F_0}{\beta}(1 + \beta\sigma W_T) - \frac{F_0}{\beta} + F_0 = F_0 + \sigma F_0 W_T$ (Bachelier with $\sigma_N = \sigma F_0$). Lower bound: $F_T > -\frac{1-\beta}{\beta}F_0$.

1.6 Pricing displaced diffusion with the Black formula [S11–S12]

The idea in words
Compare the two solutions side by side: the DD forward is literally a Black forward (start F_0/β, vol $\beta\sigma$) minus a constant shift. A call on "Black forward minus shift" with strike K is the same as a call on the Black forward with strike K+shift. So the existing BlackCall function prices it with modified inputs; no new formula needed.

Slide question [S11]: given

$$\text{Black: } F_T = F_0 e^{-\frac{\sigma^2 T}{2} + \sigma W_T}, \qquad \text{DD: } F_T = \frac{F_0}{\beta} e^{-\frac{\beta^2\sigma^2 T}{2} + \beta\sigma W_T} - \frac{1-\beta}{\beta}F_0,$$

can BlackCall(F, K, σ, T) price a DD call? Yes:

Moving the shift onto the strike [S12]

$$\begin{aligned}
(F_T - K)^+ &= \Big(\underbrace{\tfrac{F_0}{\beta} e^{-\frac{1}{2}(\beta\sigma)^2 T + \beta\sigma W_T}}_{\text{Black forward: start } F_0/\beta,\ \text{vol } \beta\sigma} - \Big[K + \tfrac{1-\beta}{\beta}F_0\Big]\Big)^+ && \text{(group the constants)}\\
\text{DD call} &= \text{Black}\Big(\frac{F_0}{\beta},\ K + \frac{1-\beta}{\beta}F_0,\ \sigma\beta,\ T\Big) && \text{(same payoff, same law)}
\end{aligned}$$

where

F_0/β	"forward" input to Black.
$K + \frac{1-\beta}{\beta}F_0$	"strike" input: real strike plus the displacement.
$\sigma\beta$	"volatility" input.
Black(F, K, σ, T)	$D[F\Phi(d_1) - K\Phi(d_2)]$, $d_{1,2} = \frac{\log(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$, D the discount factor.

[S12]: written with W_t^*, a Brownian motion under the pricing (forward) measure, so F_t is driftless.

1.7 Fitting with displaced diffusion; why vol must be random [S13–S14]

The idea in words
DD's one extra parameter β tilts the implied-vol curve between flat (lognormal) and steep (normal), so it can match the overall slope. But it is still a *local* model with deterministic volatility: it cannot produce the curvature (wings lifting on both sides). Look at real return data: periods of calm and periods of turbulence cluster together. Constant-vol models produce i.i.d. returns with no clustering. The natural fix is to let volatility be random.

- Choosing β (0.2–0.8 shown) gives a closer fit than either pure model [S13].
- Still not accurate: DD gives a monotone skew and cannot match the curvature (upturn at high strikes) [S13].

- **Which returns are real? [S14]** (answered here): the series showing *volatility clustering* (calm periods, turbulent periods such as 2008, big spikes, fat tails). The other has constant variance (i.i.d. normal), which is what Black–Scholes assumes.

1.8 Stochastic volatility [S15]

The idea in words
Give volatility its own SDE. The simplest choice: a driftless lognormal process, the same SDE as a GBM with zero drift. So the solution is the familiar exponential martingale: volatility stays positive, its expected value stays at σ_0, and ν (vol-of-vol) says how wildly it moves. Random vol mixes many normal distributions with different widths, and a mixture of normals has fatter tails than a single normal: that is exactly what lifts the wings of the smile.

$$\mathrm{d}\sigma_t = \nu\sigma_t\,\mathrm{d}W_t^{\sigma}.$$

where

σ_t	volatility at time t: now a random process.
ν	volatility of volatility.
W_t^{σ}	Brownian motion driving the volatility (may be correlated with the price's).

Solving the SDE [S15] (same as GBM with zero drift)

$$\mathrm{d}\log\sigma_t = \frac{1}{\sigma_t}\,\mathrm{d}\sigma_t - \frac{1}{2}\frac{1}{\sigma_t^2}(\mathrm{d}\sigma_t)^2 = \nu\,\mathrm{d}W_t^{\sigma} - \tfrac{1}{2}\nu^2\,\mathrm{d}t \qquad \text{(Itô on log; } (\mathrm{d}\sigma)^2 = \nu^2\sigma^2\,\mathrm{d}t\text{)}$$

$$\sigma_T = \sigma_0\exp\Big(-\tfrac{1}{2}\nu^2T + \nu W_T^{\sigma}\Big) = \sigma_0\exp\Big(-\tfrac{1}{2}\nu^2T + \nu\sqrt{T}\,\mathcal{N}(0,1)\Big) \qquad \text{(integrate, exponentiate)}$$

$$\mathbb{E}[\sigma_T] = \sigma_0 e^{-\frac{1}{2}\nu^2T}e^{\frac{1}{2}\nu^2T} = \sigma_0 \qquad \text{(normal MGF; driftless)}$$

1.9 Heston model [S16]

The idea in words
Heston (1993) models the *variance* V_t as a mean-reverting process (like Vasicek, Session 5) with a $\sqrt{V}$ in the noise (CIR), so the noise dies out as $V \to 0$ and variance stays non-negative. Volatility shocks are correlated with price shocks; for equities $\rho < 0$ (prices fall, vol spikes: leverage effect), which produces the downside skew. Vanilla prices need a numerical one-dimensional integral (Fourier).

$$\begin{cases} \mathrm{d}S_t = rS_t\,\mathrm{d}t + \sqrt{V_t}S_t\,\mathrm{d}W_t^S \\ \mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \nu\sqrt{V_t}\,\mathrm{d}W_t^V \end{cases} \qquad \mathrm{d}W_t^S\,\mathrm{d}W_t^V = \rho\,\mathrm{d}t.$$

where

V_t	instantaneous variance; $\sqrt{V_t}$ is the volatility.
κ, θ	speed of mean reversion; long-run variance.
$\nu\sqrt{V_t}\,\mathrm{d}W^V$	square-root diffusion (CIR): keeps $V \geq 0$.
ρ	correlation of stock and variance shocks; $\rho < 0$ for equities gives the downward skew *[Beyond slides]*.

Vanilla European prices: 1-d integral evaluated numerically [S16]; popular with equity desks [S16]. *[Beyond slides]* Feller condition $2\kappa\theta > \nu^2$ keeps V_t strictly positive.

1.10 SABR model [S17]

The idea in words

SABR (Hagan 2002) combines three ingredients, one per smile feature: a CEV backbone F^β (how vol moves with the level; β between normal and lognormal, like DD), a driftless lognormal volatility α_t with vol-of-vol ν (curvature), and correlation ρ between forward and vol shocks (skew). Its big practical advantage: Hagan's asymptotic formula gives the Black implied vol directly, so pricing a vanilla is "compute σ_{SABR}, plug into Black".

$$\begin{cases} \mathrm{d}F_t = \alpha_t F_t^\beta\,\mathrm{d}W_t^F \\ \mathrm{d}\alpha_t = \nu\alpha_t\,\mathrm{d}W_t^\alpha \end{cases} \qquad \mathrm{d}W_t^F\,\mathrm{d}W_t^\alpha = \rho\,\mathrm{d}t.$$

where

α_t	stochastic volatility level; $\alpha_0 = \alpha$ today.
F_t^β	backbone (CEV): $\beta = 1$ lognormal, $\beta = 0$ normal.
ν	vol-of-vol: controls the smile's curvature.
ρ	correlation of forward and vol shocks: controls the skew (tilt).

Use [S17]: compute $\sigma_{SABR}(F_0, K, \dots)$ and plug into Black. Much faster than Heston; used across asset classes.

1.11 Hagan's SABR implied volatility [S18]

$$\sigma_{SABR}(F_0, K, \alpha, \beta, \rho, \nu) = \frac{\alpha}{(F_0K)^{\frac{1-\beta}{2}}\left\{1 + \frac{(1-\beta)^2}{24}\log^2\frac{F_0}{K} + \frac{(1-\beta)^4}{1920}\log^4\frac{F_0}{K} + \cdots\right\}} \times \frac{z}{x(z)} \times \left\{1 + \left[\frac{(1-\beta)^2}{24}\frac{\alpha^2}{(F_0K)^{1-\beta}} + \frac{1}{4}\frac{\rho\beta\nu\alpha}{(F_0K)^{\frac{1-\beta}{2}}} + \frac{2-3\rho^2}{24}\nu^2\right]T + \cdots\right\}$$

$$z = \frac{\nu}{\alpha}(F_0K)^{\frac{1-\beta}{2}}\log\frac{F_0}{K}, \qquad x(z) = \log\left[\frac{\sqrt{1-2\rho z+z^2}+z-\rho}{1-\rho}\right].$$

where

$\alpha/(F_0K)^{\frac{1-\beta}{2}}$	converts the CEV vol αF^β into a lognormal vol at the geometric-mean level $\sqrt{F_0K}$ (since $\alpha F^\beta/F = \alpha F^{\beta-1}$).
$\{1 + \frac{(1-\beta)^2}{24}\log^2\cdots\}$	backbone correction away from ATM; vanishes when $\beta = 1$.
$z/x(z)$	stochastic-vol smile factor: depends on ν, ρ; equals 1 ATM.
$[\cdots]T$	small time correction: backbone, correlation, vol-of-vol terms.

Two sanity limits *[Beyond slides]*

ATM ($K = F_0$): $z \to 0$; $x(z) \approx \log\frac{1-\rho+z(1-\rho)+\cdots}{1-\rho} \approx z$, so $z/x(z) \to 1$ and $\sigma_{ATM} \approx \frac{\alpha}{F_0^{1-\beta}}\{1 + [\cdots]T\}$ (code must treat this limit separately). $\beta = 1$, $\nu \to 0$: denominator $= \alpha$, $z/x \to 1$, bracket $\to 1$, so $\sigma_{SABR} = \alpha$: flat Black.

Practice *[Beyond slides]*: fix β (often 0.5 or by asset class), calibrate α, ρ, ν to the market smile for each expiry. Code is in the course's Jupyter notebook [S18].

1.12 Fit comparison [S19–S20]

- Displaced diffusion (best β) has too few degrees of freedom: it misses the low-strike steepness and the high-strike upturn [S19].
- SABR fits the implied volatility smile well and is quick to compute, hence popular [S20].

1.13 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
from scipy.optimize import brentq
def black_call(F, K, sigma, T, D=1.0):
    d1 = (np.log(F/K) + 0.5*sigma**2*T) / (sigma*np.sqrt(T))
    return D*(F*norm.cdf(d1) - K*norm.cdf(d1 - sigma*np.sqrt(T)))
def bachelier_call(F, K, sigma, T, D=1.0):
    d = (F - K) / (sigma*np.sqrt(T))
    return D*((F - K)*norm.cdf(d) + sigma*np.sqrt(T)*norm.pdf(d))
def implied_vol(price, F, K, T, D=1.0):
    return brentq(lambda s: black_call(F, K, s, T, D) - price, 1e-6, 5.0)
def dd_call(F, K, sigma, beta, T, D=1.0):
    """Displaced diffusion dF = sigma [beta F + (1-beta) F0] dW, priced with Black."""
    return black_call(F/beta, K + (1-beta)/beta*F, sigma*beta, T, D)
def sabr_vol(F, K, T, alpha, beta, rho, nu):
    """Hagan et al. (2002) lognormal implied vol."""
    B = lambda fk: 1 + ((1-beta)**2/24*alpha**2/fk**(1-beta)
                        + 0.25*rho*beta*nu*alpha/fk**((1-beta)/2) + (2-3*rho**2)/24*nu**2)*T
    if abs(F - K) < 1e-12:                        # ATM: z/x(z) -> 1
        return alpha/F**(1-beta)*B(F*F)
    fk, lfk = F*K, np.log(F/K)
    z = nu/alpha*fk**((1-beta)/2)*lfk
    xz = np.log((np.sqrt(1 - 2*rho*z + z**2) + z - rho) / (1 - rho))
    A = alpha / (fk**((1-beta)/2)*(1 + (1-beta)**2/24*lfk**2 + (1-beta)**4/1920*lfk**4))
    return A*(z/xz)*B(fk)
F, T = 846.9, 1.38                                 # Google example
for K in (400, 846.9, 1400):                       # normal model -> skew in Black vols
    print(K, round(implied_vol(bachelier_call(F, K, 0.26*F, T), F, K, T), 4))   # 0.3725 0.261 0.2006
print(round(dd_call(F, 900, 0.26, 0.4, T), 3))    # 80.009 (Monte Carlo: 79.963 +/- 0.05)
print(round(sabr_vol(F, 1200, T, 0.26, 1.0, 0.0, 1e-8), 6))   # 0.26: beta=1, nu->0 is flat Black
```

2 Comparison tables

2.1 Pricing vs reporting models

	Pricing model [S2]	Reporting model [S3]
Purpose	price and hedge derivatives	quote market prices as vols
Requirement	realistic dynamics, accurate Greeks	simple, analytic formula
Parameters	calibrated to liquid options	implied vols on screens
Examples	SABR, Heston, displaced diffusion	Black–Scholes, Bachelier

2.2 Models and the smile

Model	SDE	Free params	Smile in BS implied vol
Black–Scholes	$\mathrm{d}F = \sigma F\,\mathrm{d}W$	σ	flat
Bachelier	$\mathrm{d}F = \sigma_N\,\mathrm{d}W$	σ_N	fixed downward skew
Displaced diffusion	$\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\,\mathrm{d}W$	σ, β	adjustable skew, no curvature
Heston	see 1.9	$V_0, \kappa, \theta, \nu, \rho$	skew and curvature; numerical integral
SABR	see 1.10	α, β, ρ, ν	skew (ρ, β), curvature (ν); closed-form vol

2.3 What each SABR parameter does *[Beyond slides]*

α	overall level of volatility (mainly ATM vol)
β	backbone: how ATM vol moves with F; usually fixed
ρ	skew: $\rho < 0$ raises low-strike vols
ν	curvature: higher ν lifts both wings

2.4 Heston vs SABR

	Heston [S16]	SABR [S17–S18]
Random quantity	variance V_t, mean-reverting	volatility α_t, driftless lognormal
Underlying	spot with drift r	forward, driftless, CEV F^β
Vanilla pricing	1-d numerical integral	closed-form implied vol + Black
Speed / typical use	slower / equity desks	fast / all asset classes, esp. rates

3 Master formula sheet

- **Implied vol [S4]:** $C^{\text{mkt}}(K) = \text{BS}(S, K, r, \sigma_K, T)$, unique since vega > 0.
- **Black / Bachelier (forward form):** $D[F\Phi(d_1) - K\Phi(d_2)]$, $d_{1,2} = \frac{\log(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$; $D[(F-K)\Phi(d) + \sigma_N\sqrt{T}\phi(d)]$, $d = \frac{F-K}{\sigma_N\sqrt{T}}$.
- **Displaced diffusion [S10–S12]:** $F_T = \frac{F_0}{\beta} e^{-\frac{1}{2}\beta^2\sigma^2 T + \beta\sigma W_T} - \frac{1-\beta}{\beta}F_0$; call $= \text{Black}\big(\frac{F_0}{\beta}, K + \frac{1-\beta}{\beta}F_0, \sigma\beta, T\big)$.
- **Stochastic vol [S15]:** $\mathrm{d}\sigma = \nu\sigma\,\mathrm{d}W^\sigma \Rightarrow \sigma_T = \sigma_0 e^{-\frac{1}{2}\nu^2 T + \nu W_T^\sigma}$.
- **Heston [S16]:** $\mathrm{d}S = rS\,\mathrm{d}t + \sqrt{V}S\,\mathrm{d}W^S$, $\mathrm{d}V = \kappa(\theta - V)\,\mathrm{d}t + \nu\sqrt{V}\,\mathrm{d}W^V$, $\mathrm{d}W^S\,\mathrm{d}W^V = \rho\,\mathrm{d}t$.
- **SABR [S17–S18]:** $\mathrm{d}F = \alpha F^\beta\,\mathrm{d}W^F$, $\mathrm{d}\alpha = \nu\alpha\,\mathrm{d}W^\alpha$, ρ; price $= \text{Black}(F_0, K, \sigma_{SABR}, T)$.

4 Glossary

- **ATM / ITM / OTM:** at / in / out of the money [S7]. **Backbone:** how ATM vol moves with the forward (SABR β) *[Beyond slides]*.
- **Calibration:** choosing model parameters to fit market prices [S2].
- **Displaced diffusion (shifted lognormal):** $\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\,\mathrm{d}W$ [S10].
- **Implied volatility:** the σ that makes Black–Scholes match a market price [S4].
- **Leverage effect:** negative correlation between price and volatility moves *[Beyond slides]*.
- **Pricing / reporting model:** used to price and hedge / converts prices to implied vols [S2–S3].
- **Skew / smile:** implied vol varying with strike [S5]. **Vol-of-vol ν:** volatility of the volatility process [S15].
- **Volatility clustering:** large moves follow large moves; calm follows calm [S14].

5 Self-test

Questions

Q1. ([S2–S3]) Contrast pricing and reporting models. Why are implied vols quoted rather than prices?

Q2. ([S4]) Define implied volatility. Why is it unique?

Q3. ([S5–S7]) Describe the equity smile, when it appeared and why.

Q4. ([S8–S9]) Why can't Black–Scholes or Bachelier fit the smile? What shape does Bachelier produce in Black implied vol?

Q5. ([S10]) Solve $\mathrm{d}F = \sigma[\beta F + (1-\beta)F_0]\,\mathrm{d}W$. What are the limits $\beta = 1$ and $\beta \to 0$?

Q6. ([S11–S12]) Price a DD call with BlackCall. Give the inputs.

Q7. ([S13–S14]) Why is DD not enough, and what evidence shows vol is random?

Q8. ([S15]) Solve $\mathrm{d}\sigma = \nu\sigma\,\mathrm{d}W$ and find $\mathbb{E}[\sigma_T]$.

Q9. ([S16]) Write the Heston SDEs and explain each parameter.

Q10. ([S17–S18]) Write the SABR SDEs, state what ρ and ν control, and explain how SABR prices a vanilla.

Solutions

S1. See 1.1 and table 2.1. Vols normalise for spot, strike, expiry and rates, so they are comparable and more stable.

S2. The σ_K with $\mathrm{BS}(S, K, r, \sigma_K, T) = C^{\mathrm{mkt}}(K)$; unique because price is strictly increasing in σ.

S3. Low strikes have higher implied vol (downward skew), steeper for short expiries; appeared after the 1987 crash; demand for OTM put protection.

S4. One parameter fits only ATM. Bachelier gives Black vols falling with strike (0.37 / 0.26 / 0.20 at $K = 400$ / 846.9 / 1400).

S5. See 1.5. $\beta = 1$: Black. $\beta \to 0$: $F_T \to F_0 + \sigma F_0 W_T$, Bachelier with $\sigma_N = \sigma F_0$.

S6. $\mathrm{Black}(F_0/\beta,\ K + \frac{1-\beta}{\beta}F_0,\ \sigma\beta,\ T)$.

S7. One extra parameter only tilts the skew; no curvature. Real returns show volatility clustering and fat tails.

S8. $\sigma_T = \sigma_0 e^{-\frac{1}{2}\nu^2 T + \nu W_T}$; $\mathbb{E}[\sigma_T] = \sigma_0$.

S9. See 1.9.

S10. See 1.10. ρ = skew; ν = curvature. Compute $\sigma_{SABR}(F_0, K)$ with Hagan's formula, then Black.

Appendix A. Slide coverage checklist

Slide	Content	Section
S2–S3	Pricing models; reporting models	1.1
S4	Implied volatility per strike	1.2
S5–S7	Smile before and after 1987; demand for protection; by expiry	1.3
S8–S9	Fitting with one vol parameter; Google example	1.4
S10	Displaced diffusion SDE and solution	1.5
S11–S12	Pricing DD with BlackCall	1.6
S13–S14	DD fit; real vs simulated returns	1.7
S15	Stochastic volatility	1.8
S16	Heston	1.9
S17–S18	SABR; Hagan's formula	1.10–1.11
S19–S20	DD vs SABR fit	1.12

Appendix B. Flag summary (△!)

1. [S10, S17] β is used for two different parameters (DD weight and SABR exponent).
2. [S11–S12] The BlackCall signature omits the discount factor; multiply by $D = e^{-rT}$ when F is a forward.
3. [S16] Heston's ν is the volatility of variance, unlike SABR's ν (volatility of volatility).
4. [S18] The formula has a removable singularity at $K = F_0$; implementations must treat ATM separately.
5. [S3, S10, S11, S13, S14] Questions posed on the slides are answered here with my own working.